

Sakash Srivastava

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SUMMARY

Shipped two LLM systems solo this year, one running live in production on Azure with Docker, CI/CD, authentication and a full test suite. Work centres on agentic architectures, retrieval, and pulling structured data out of unstructured documents reliably enough to act on. Comfortable taking an idea from prototype to something people can use.

TECHNICAL SKILLS

LLM & AI: LLMs (GPT, Claude), Prompt Engineering, RAG, Agentic Systems, Tool Calling, Structured Outputs, Embeddings, Vector Search, Transformers

Languages: Python, C++, SQL

Frameworks & APIs: LangChain, LangGraph, Flask, REST APIs, Pydantic

Data: SQLite, ChromaDB, Pandas, NumPy, OCR (Tesseract, EasyOCR)

ML & Deployment: Scikit-Learn, Docker, Azure, GitHub Actions CI/CD

EXPERIENCE

King's College London

Visiting Research Intern

Jun 2026 – Jul 2026

London, United Kingdom

- Worked with PhD researchers on orbital wall segmentation from MRI, a clinically important and previously unsolved problem, contributing across the full pipeline from data generation to model evaluation.
- Built a CT-to-MRI registration pipeline to auto-generate bone labels, cutting annotation from hours per case to minutes across 40+ paired cases and establishing a 0.938 mean Dice soft-tissue baseline across 9 structures.
- Benchmarked the wall model across 8 architecture and loss-function configurations using boundary-aware metrics (HD95, ASSD) suited to thin structures.

Netaji Subhas University of Technology

Research Intern

Jun 2025 – Jul 2025

New Delhi, India

- Extended the TLcR-RL patch-wise super-resolution framework, replacing fixed-K neighbour selection with a PSNR-driven adaptive-K scheme evaluated per image patch.
- Benchmarked adaptive-K against the fixed-K baseline (33.63 PSNR / 0.930 SSIM), diagnosed why naive adaptation underperformed, and prototyped an ANFIS model in MATLAB for dynamic K selection.

UpValue Tech Pvt. Ltd.

Machine Learning Intern (Remote)

Jan 2025 – May 2025

Jaipur, Rajasthan

- Built an end-to-end counterfeit verification pipeline in Python and OpenCV covering background removal, CNN feature extraction, dual-engine OCR (Tesseract, EasyOCR) and stitching density analysis across 6 product viewpoints.
- Benchmarked 6 model families across 9 classification and similarity metrics; the custom 3-block CNN outperformed all baselines and was selected as the production architecture.
- Designed a priority-weighted decision engine combining text, similarity and build-quality signals into 3 verdicts, with Grad-CAM heatmaps so platform moderators could interpret every flag.

PROJECTS

SegLit: Agentic Research Assistant | *Python, LLMs, Flask, Docker, Azure, ChromaDB, SQLite*

Aug 2026

- Built an agentic RAG system over 276 papers: layout-aware ingestion of 2,435 two-column PDF pages and 732 result tables into 8,139 embeddings across ChromaDB and BM25, answered by a hand-written tool-calling agent returning cited comparative answers.
- Extracted reported results into a verified SQLite metrics table (829 rows, 267 architectures, 72 datasets) with a verbatim source-match check that discards 5% hallucinated values.
- Benchmarked retrieval on a hand-labelled 52-question set (recall@5, MRR), showing dense embeddings outperform BM25 and cross-encoder reranking on table-heavy data.
- Shipped a secure multi-user Flask app (auth, per-user history, bring-your-own-key, semantic answer cache serving repeats at 0 tokens) with CSRF protection, rate limiting and a 26-check integration test suite, deployed live over HTTPS via Docker on Azure with GitHub Actions CI/CD.
- GitHub Link: SegLit | Live Website Link: SegLit-Assistant

Multi-Agent Orchestration System | *Python, LLMs, Pydantic, SQLite, Streamlit, Docker*

Jul 2026

- Built a multi-agent AI system from scratch without LangGraph or CrewAI: a supervisor agent decomposes tasks into dependency-ordered plans, specialist agents execute with sandboxed tools, and an LLM-as-judge reviewer validates every output against system-verified tool logs.

- Cut the reviewer’s false-rejection rate from 5/6 to 0 by supplying verified tool-call evidence; engineered self-correcting structured outputs (Pydantic), SQLite-backed state making crash recovery and human-in-the-loop approval the same pause/resume mechanism, and per-call cost and latency telemetry.
- Diagnosed and fixed 6 documented production-class failures; shipped with a 19-test suite including negative tests for sandbox escapes, Docker packaging and design documentation.
- GitHub Link: Multi-Agent Orchestration System

Real Estate Price Prediction | *Python, Flask, XGBoost, HTML/CSS*

Jun 2025

- Built and deployed a Flask web app serving real-time price predictions on 13,000+ Bengaluru listings, covering data cleaning, feature engineering, XGBoost training (80%+ R^2) and Isolation Forest outlier detection.
- GitHub Link: Real Estate Price Predictor

EDUCATION

The LNM Institute of Information Technology
B.Tech in Electronics and Communication Engineering, CGPA: 7.60/10

Aug 2023 – May 2027
Jaipur, Rajasthan

CERTIFICATIONS & ACHIEVEMENTS

Prompt Engineering & Programming with OpenAI, Columbia Plus (Prompt Engineering, OpenAI API, LLMs, LangChain)
Machine Learning in Production (ML lifecycle, deployment, monitoring, error analysis)
Databases and SQL for Data Science with Python (SQL, relational databases, SQLite, Python-SQL integration)
Supervised Machine Learning: Regression and Classification, Coursera (Stanford)
Advanced Learning Algorithms, Coursera
Top 5% at Adobe India Hackathon 2025

POSITIONS OF RESPONSIBILITY

Training, Placements & Corporate Relations Cell, LNMIIT
Senior Executive

- Coordinate company assessments and candidate onboarding for campus recruitment.

TEDxLNMIIT
Organizer

- Handled event management, sponsorships, speaker outreach and content creation.

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